

quant1c

June 8, 2026

```
[46]: # this is my notebook for quant 1c project. looking at before and after eu
      ↪tethered plastic bottle cap regulation
```

```
[47]: import pandas as pd
      import numpy as np
      import scipy.stats as sp
```

```
[48]: austria = pd.read_csv("./data/detailed-summary-austria.csv")
      belgium = pd.read_csv("./data/detailed-summary-belgium.csv")
      bulgaria = pd.read_csv("./data/detailed-summary-bulgaria.csv")
      croatia = pd.read_csv("./data/detailed-summary-croatia.csv")
      cyprus = pd.read_csv("./data/detailed-summary-cyprus.csv")
      czechia = pd.read_csv("./data/detailed-summary-czechia.csv")
      denmark = pd.read_csv("./data/detailed-summary-denmark.csv")
      estonia = pd.read_csv("./data/detailed-summary-estonia.csv")
      finland = pd.read_csv("./data/detailed-summary-finland.csv")
      france = pd.read_csv("./data/detailed-summary-france.csv")
      germany = pd.read_csv("./data/detailed-summary-germany.csv")
      greece = pd.read_csv("./data/detailed-summary-greece.csv")
      hungary = pd.read_csv("./data/detailed-summary-hungary.csv")
      ireland = pd.read_csv("./data/detailed-summary-ireland.csv")
      italy = pd.read_csv("./data/detailed-summary-italy.csv")
      latvia = pd.read_csv("./data/detailed-summary-latvia.csv")
      lithuania = pd.read_csv("./data/detailed-summary-lithuania.csv")
      malta = pd.read_csv("./data/detailed-summary-malta.csv")
      netherlands = pd.read_csv("./data/detailed-summary-netherlands.csv")
      poland = pd.read_csv("./data/detailed-summary-poland.csv")
      portugal = pd.read_csv("./data/detailed-summary-portugal.csv")
      romania = pd.read_csv("./data/detailed-summary-romania.csv")
      slovakia = pd.read_csv("./data/detailed-summary-slovakia.csv")
      slovenia = pd.read_csv("./data/detailed-summary-slovenia.csv")
      spain = pd.read_csv("./data/detailed-summary-spain.csv")
      sweden = pd.read_csv("./data/detailed-summary-sweden.csv")

      df = pd.concat([
          austria,
          belgium,
```

```

bulgaria,
croatia,
cyprus,
czechia,
denmark,
estonia,
finland,
france,
germany,
greece,
hungary,
ireland,
italy,
latvia,
lithuania,
malta,
netherlands,
poland,
portugal,
romania,
slovakia,
slovenia,
spain,
sweden
], ignore_index=True)

df

```

```

[48]:
      Cleanup ID      Zone \
0      282231  Schwaz District, Austria
1      282232  Schwaz District, Austria
2      282354      Vienna, Austria
3      305521      Vienna, Austria
4      Totals:      NaN
...
5946    364768  Ängelholm Municipality, Sweden
5947    364769  Österåker Municipality, Sweden
5948    364770      Varberg Municipality, Sweden
5949    393537  Robertsfors Municipality, Sweden
5950    Totals:      NaN

      State Country  Latitude  Longitude Cleanup Type \
0      Tyrol, Austria  Austria  47.166778  11.865272      land
1      Tyrol, Austria  Austria  47.166707  11.865285      land
2      Vienna, Austria  Austria  48.194622  16.446747      land
3      Vienna, Austria  Austria  48.215486  16.400839      land
4      NaN      NaN      NaN      NaN      NaN

```

...	...	...	...	...	...
5946	Skåne County, Sweden	Sweden	56.283300	12.733300	land
5947	Stockholm County, Sweden	Sweden	59.456700	18.358951	land
5948	Halland County, Sweden	Sweden	57.100000	12.250000	land
5949	Västerbotten County, Sweden	Sweden	63.991349	20.872819	land
5950		NaN	NaN	NaN	NaN

	Environment Cleanup Date		Group Name \
0	NaN	2023-05-31	NaN
1	NaN	2023-05-31	NaN
2	NaN	2023-05-29	NaN
3	inland	2023-09-16	AKTI Prater Park
4	NaN	NaN	NaN
...	...	...	...
5946	saltwater	2024-09-21	Vejbystrand
5947	saltwater	2024-09-21	Brevik
5948	saltwater	2024-09-21	Kustomrde frn Taparken till Turisttoaletterna
5949	saltwater	2025-09-20	Clean Sweden
5950	NaN	NaN	NaN

	...	Other plastic/foam packaging	Tobacco packaging/wrap	Cigar tips \
0	...	NaN	NaN	NaN
1	...	NaN	NaN	NaN
2	...	NaN	NaN	NaN
3	...	NaN	NaN	NaN
4	...	0.0	0.0	0.0
...	...	...	...	...
5946	...	NaN	NaN	NaN
5947	...	NaN	NaN	NaN
5948	...	NaN	NaN	NaN
5949	...	NaN	NaN	NaN
5950	...	0.0	0.0	0.0

	Cigarette lighters	Fireworks	Other tobacco (packaging, lighter, etc.) \
0	NaN	NaN	NaN
1	NaN	NaN	NaN
2	NaN	NaN	NaN
3	NaN	NaN	NaN
4	0.0	0.0	0.0
...	...	...	...
5946	NaN	NaN	NaN
5947	NaN	NaN	NaN
5948	NaN	NaN	NaN
5949	NaN	NaN	NaN
5950	0.0	0.0	10.0

Foam pieces Glass pieces Plastic pieces Total Items Collected

0	NaN	NaN	NaN	16.0
1	NaN	NaN	NaN	5.0
2	NaN	NaN	NaN	13.0
3	NaN	NaN	NaN	NaN
4	0.0	0.0	0.0	NaN
...	...	...	...	...
5946	NaN	NaN	NaN	2.0
5947	NaN	NaN	NaN	2.0
5948	NaN	NaN	NaN	1.0
5949	NaN	NaN	NaN	NaN
5950	0.0	0.0	13.0	NaN

[5951 rows x 75 columns]

```
[49]: df = df.drop(columns=[
    'Zone', 'State', 'Latitude', 'Longitude',
    'Cleanup Type', 'Environment', 'Group Name',
    'Adults', 'Children',
    'Distance mi', 'Area mi', '# of bags',
    'Grocery bags (plastic)', 'Other bags (plastic)',
    'Beverage bottles (glass)', 'Beverage cans',
    'Beverage sachets/pouches', 'Bottle caps (metal)',
    'Cigarette butts', 'Cups, Plates (foam)',
    'Cups, Plates (paper)', 'Cups, Plates (plastic)',
    'Food containers (foam)', 'Food containers (plastic)',
    'Food wrappers (candy, chips, etc.)',
    'Lids (plastic)', 'Straws/stirrers (plastic)',
    'Utensils (plastic)',
    'Lines, nets, traps, ropes, etc.', 'Foam dock pieces',
    'Appliances (refrigerators, washers, etc.)',
    'Construction materials', 'Tires', '6-Pack holders',
    'Foam packaging',
    'Other plastic bottles (oil, bleach, etc.)',
    'Strapping bands', 'Balloons', 'Clothing',
    'E-cigarettes',
    'Electronic waste (phones, batteries)',
    'Footwear (shoes/slippers)', 'Paper bags',
    'Tobacco products (lighters, cigar tips, wrap)',
    'Toys', 'Other plastic waste',
    'Other waste (metal, paper, etc.)',
    'Condoms', 'Cotton bud sticks (swabs)',
    'Diapers', 'Gloves & masks (PPE)',
    'Syringes', 'Tampons & applicators',
    'Plastic/foam pieces',
    'Fishing buoys, pots & traps',
    'Fishing net & pieces',
    'Fishing line (1 yard/meter = 1 piece)',
```

```

'Rope (1 yard/meter = 1 piece)',
'Other plastic/foam packaging',
'Tobacco packaging/wrap', 'Cigar tips',
'Cigarette lighters', 'Fireworks',
'Other tobacco (packaging, lighter, etc.)',
'Foam pieces', 'Glass pieces', 'Plastic pieces'
])

```

[50]: df

```

[50]:
Cleanup ID Country Cleanup Date People Pounds \
0      282231 Austria 2023-05-31 1.0 0.24
1      282232 Austria 2023-05-31 1.0 0.37
2      282354 Austria 2023-05-29 1.0 0.21
3      305521 Austria 2023-09-16 2.0 3.31
4      Totals:      NaN      NaN      NaN      NaN
...
5946    364768 Sweden 2024-09-21 28.0 120.00
5947    364769 Sweden 2024-09-21 21.0 120.00
5948    364770 Sweden 2024-09-21 28.0 90.00
5949    393537 Sweden 2025-09-20 2962.0 17,920.00
5950    Totals:      NaN      NaN      NaN      NaN

Beverage bottles (plastic) Bottle caps (plastic) Total Items Collected
0                          3.0                2.0                16.0
1                          NaN                NaN                 5.0
2                          NaN                NaN                13.0
3                          NaN                NaN                NaN
4                          3.0                2.0                NaN
...
5946                      NaN                NaN                 2.0
5947                      NaN                NaN                 2.0
5948                      NaN                NaN                 1.0
5949                      NaN                NaN                NaN
5950                      52.0               27.0                NaN

```

[5951 rows x 8 columns]

```

[51]: #get rid of totals because will do manually with specific time frames
# also getting rid of rows with 0 if there is any because dividing by 0 is bad.
df = df[df["Cleanup ID"] != "Totals:"]
df = df[df["Beverage bottles (plastic)"] != 0]
df = df[df["Bottle caps (plastic)"] != 0]

df

```

```
[51]:
```

	Cleanup ID	Country	Cleanup Date	People	Pounds	\
0	282231	Austria	2023-05-31	1.0	0.24	
1	282232	Austria	2023-05-31	1.0	0.37	
2	282354	Austria	2023-05-29	1.0	0.21	
3	305521	Austria	2023-09-16	2.0	3.31	
5	162682	Belgium	2021-03-23	1.0	2.00	
...	...	...	...	...		
5945	364767	Sweden	2024-09-21	27.0	80.00	
5946	364768	Sweden	2024-09-21	28.0	120.00	
5947	364769	Sweden	2024-09-21	21.0	120.00	
5948	364770	Sweden	2024-09-21	28.0	90.00	
5949	393537	Sweden	2025-09-20	2962.0	17,920.00	

	Beverage bottles (plastic)	Bottle caps (plastic)	Total Items Collected
0	3.0	2.0	16.0
1	NaN	NaN	5.0
2	NaN	NaN	13.0
3	NaN	NaN	NaN
5	NaN	NaN	NaN
...	...	...	...
5945	NaN	NaN	2.0
5946	NaN	NaN	2.0
5947	NaN	NaN	2.0
5948	NaN	NaN	1.0
5949	NaN	NaN	NaN

[5925 rows x 8 columns]

```
[52]: clean_df = df.copy()
```

```
[53]: clean_df['cap_bottle_ratio'] = (clean_df['Bottle caps (plastic)'] /
↳ clean_df['Beverage bottles (plastic)'])
clean_df #need to add this column for later
```

```
[53]:
```

	Cleanup ID	Country	Cleanup Date	People	Pounds	\
0	282231	Austria	2023-05-31	1.0	0.24	
1	282232	Austria	2023-05-31	1.0	0.37	
2	282354	Austria	2023-05-29	1.0	0.21	
3	305521	Austria	2023-09-16	2.0	3.31	
5	162682	Belgium	2021-03-23	1.0	2.00	
...	...	...	...	...		
5945	364767	Sweden	2024-09-21	27.0	80.00	
5946	364768	Sweden	2024-09-21	28.0	120.00	
5947	364769	Sweden	2024-09-21	21.0	120.00	
5948	364770	Sweden	2024-09-21	28.0	90.00	
5949	393537	Sweden	2025-09-20	2962.0	17,920.00	

	Beverage bottles (plastic)	Bottle caps (plastic)	\
0	3.0		2.0
1	NaN		NaN
2	NaN		NaN
3	NaN		NaN
5	NaN		NaN
...	...		...
5945	NaN		NaN
5946	NaN		NaN
5947	NaN		NaN
5948	NaN		NaN
5949	NaN		NaN

	Total Items Collected	cap_bottle_ratio
0	16.0	0.666667
1	5.0	NaN
2	13.0	NaN
3	NaN	NaN
5	NaN	NaN
...	...	...
5945	2.0	NaN
5946	2.0	NaN
5947	2.0	NaN
5948	1.0	NaN
5949	NaN	NaN

[5925 rows x 9 columns]

[]:

[54]: *#now to seperate timelines*

```
[55]: clean_df['Cleanup Date'] = pd.to_datetime(clean_df['Cleanup Date'])

before = clean_df[(clean_df['Cleanup Date'] >= '2021-01-01') &
↳(clean_df['Cleanup Date'] < '2022-12-31')]
during = clean_df[(clean_df['Cleanup Date'] >= '2023-01-01') &
↳(clean_df['Cleanup Date'] < '2024-06-01')]
after = clean_df[(clean_df['Cleanup Date'] >= '2024-07-01') &
↳(clean_df['Cleanup Date'] <= '2026-05-15')]
```

[56]: before.describe()

	Cleanup Date	People	Beverage bottles (plastic)	\
count	2808	2803.000000	1393.000000	
mean	2022-01-08 23:01:01.538461696	17.398145	22.522613	
min	2021-01-01 00:00:00	1.000000	1.000000	

25%	2021-08-26 00:00:00	1.000000	2.000000
50%	2021-10-16 00:00:00	3.000000	6.000000
75%	2022-08-31 00:00:00	12.000000	20.000000
max	2022-12-29 00:00:00	9060.000000	1250.000000
std	NaN	176.069125	60.113384

	Bottle caps (plastic)	Total Items Collected	cap_bottle_ratio
count	1333.000000	2108.000000	1112.000000
mean	35.561140	510.131404	3.500580
min	1.000000	1.000000	0.010526
25%	3.000000	32.750000	0.623922
50%	10.000000	155.000000	1.333333
75%	28.000000	568.500000	2.851687
max	1500.000000	26949.000000	145.000000
std	102.781863	1183.719496	8.954448

```
[57]: during.describe()
```

```
[57]:
```

	Cleanup Date	People	Beverage bottles (plastic) \
count	1745	1745.000000	756.000000
mean	2023-09-03 19:12:00.000000	256	13.793696
min	2023-01-01 00:00:00	1.000000	1.000000
25%	2023-06-05 00:00:00	2.000000	3.000000
50%	2023-09-16 00:00:00	6.000000	8.000000
75%	2023-10-21 00:00:00	16.000000	23.000000
max	2024-05-31 00:00:00	1000.000000	910.000000
std	NaN	33.302424	77.422337

	Bottle caps (plastic)	Total Items Collected	cap_bottle_ratio
count	752.000000	1136.000000	623.000000
mean	32.819149	898.250880	4.606363
min	1.000000	1.000000	0.004657
25%	4.000000	54.000000	0.533889
50%	12.000000	241.500000	1.500000
75%	31.000000	960.000000	4.000000
max	694.000000	72913.000000	137.500000
std	69.404691	2721.043058	10.902940

```
[58]: after.describe()
```

```
[58]:
```

	Cleanup Date	People \
count	1282	1282.000000
mean	2025-03-07 08:47:55.507020	288
min	2024-07-01 00:00:00	1.000000
25%	2024-09-20 00:00:00	2.000000
50%	2024-11-16 00:00:00	5.000000
75%	2025-09-01 00:00:00	14.000000


```
max          2026-05-15 00:00:00 16292.000000
std          NaN      487.312890
```

	Beverage bottles (plastic)	Bottle caps (plastic) \
count	655.000000	639.000000
mean	19.325191	25.453834
min	1.000000	1.000000
25%	2.000000	3.000000
50%	5.000000	7.000000
75%	13.000000	17.000000
max	1847.000000	2249.000000
std	94.714295	102.333774

	Total Items Collected	cap_bottle_ratio
count	1075.000000	517.000000
mean	741.766512	3.320518
min	1.000000	0.008333
25%	40.500000	0.571429
50%	162.000000	1.217650
75%	453.500000	3.083333
max	193268.000000	154.000000
std	6031.034300	8.633378

```
[ ]:
```

```
[59]: before['Beverage bottles (plastic)'].sum()
```

```
[59]: np.float64(31374.0)
```

```
[60]: during['Beverage bottles (plastic)'].sum()
```

```
[60]: np.float64(21268.0)
```

```
[61]: after['Beverage bottles (plastic)'].sum()
```

```
[61]: np.float64(12658.0)
```

```
[62]: before['Bottle caps (plastic)'].sum()
```

```
[62]: np.float64(47403.0)
```

```
[63]: during['Bottle caps (plastic)'].sum()
```

```
[63]: np.float64(24680.0)
```

```
[64]: after['Bottle caps (plastic)'].sum()
```

```
[64]: np.float64(16265.0)
```

```
[65]: # bottle to cap ratio
before_ratio = before['Bottle caps (plastic)'].sum() / before['Beverage bottles_
↳(plastic)'].sum()
before_ratio
```

```
[65]: np.float64(1.5109007458405048)
```

```
[66]: during_ratio = during['Bottle caps (plastic)'].sum() / during['Beverage bottles_
↳(plastic)'].sum()
during_ratio
```

```
[66]: np.float64(1.1604288132405491)
```

```
[67]: after_ratio = after['Bottle caps (plastic)'].sum() / after['Beverage bottles_
↳(plastic)'].sum()
after_ratio
```

```
[67]: np.float64(1.2849581292463264)
```

```
[68]: #cpr_before = before['cap_bottle_ratio'].mean()
#cpr_before
```

```
[69]: #cpr_after = after['cap_bottle_ratio'].mean()
#cpr_after
```

```
[70]: #ttest = sp.ttest_ind(cpr_before, cpr_after)
#ttest
```

```
[71]: #ttest = sp.ttest_ind(before_ratio, after_ratio)
#ttest
```

```
[72]: cpr_before = pd.DataFrame(before["cap_bottle_ratio"])
cpr_after = pd.DataFrame(after["cap_bottle_ratio"])
#cpr is cap bottle ratio
```

```
[73]: cpr_before.describe()
```

```
[73]:      cap_bottle_ratio
count      1112.000000
mean         3.500580
std          8.954448
min          0.010526
25%          0.623922
50%          1.333333
75%          2.851687
```

max 145.000000

```
[74]: #ttest = sp.ttest_ind(before['cap_bottle_ratio'], after['cap_bottle_ratio'])  
#ttest
```

```
[75]: #apparently cant have nans in the group
```

```
[76]: clean_before = before['cap_bottle_ratio'].dropna()  
clean_after = after['cap_bottle_ratio'].dropna()
```

```
[77]: ttest = sp.ttest_ind(clean_before, clean_after)  
ttest
```

```
[77]: TtestResult(statistic=np.float64(0.3820552979761666),  
pvalue=np.float64(0.7024702431940862), df=np.float64(1627.0))
```

```
[78]: #success i think. that p value looks terrible
```

```
[79]: #lollipop visual for difference in eu per country - didnt know how to do that  
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
[80]: #have to standardise the time time periods of the ooverall clean up
```

```
[81]: clean_df['Cleanup Date'] = pd.to_datetime(clean_df['Cleanup Date'])  
  
before_1yr = clean_df[(clean_df['Cleanup Date'] >= '2022-01-01') &_  
    ↳(clean_df['Cleanup Date'] <= '2022-12-31')]  
during_1yr = clean_df[(clean_df['Cleanup Date'] >= '2023-01-01') &_  
    ↳(clean_df['Cleanup Date'] <= '2023-12-31')]  
twentyfour = clean_df[(clean_df['Cleanup Date'] >= '2024-01-01') &_  
    ↳(clean_df['Cleanup Date'] <= '2024-12-31')]  
after_1yr = clean_df[(clean_df['Cleanup Date'] >= '2025-01-01') &_  
    ↳(clean_df['Cleanup Date'] <= '2025-12-31')]
```

```
[82]: twentyfour['Beverage bottles (plastic)'].sum()
```

```
[82]: np.float64(12412.0)
```

```
[83]: categories = ['Before Regulation, 2022', 'During Transition, 2023', '2024',_  
    ↳'After Regulation, 2025']  
bottles_sum = [before_1yr['Beverage bottles (plastic)'].sum(),_  
    ↳during_1yr['Beverage bottles (plastic)'].sum(), 12412, after_1yr['Beverage_  
    ↳bottles (plastic)'].sum()]  
caps_sum = [before_1yr['Bottle caps (plastic)'].sum(), during_1yr['Bottle caps_  
    ↳(plastic)'].sum(), 15470, after_1yr['Bottle caps (plastic)'].sum()]
```

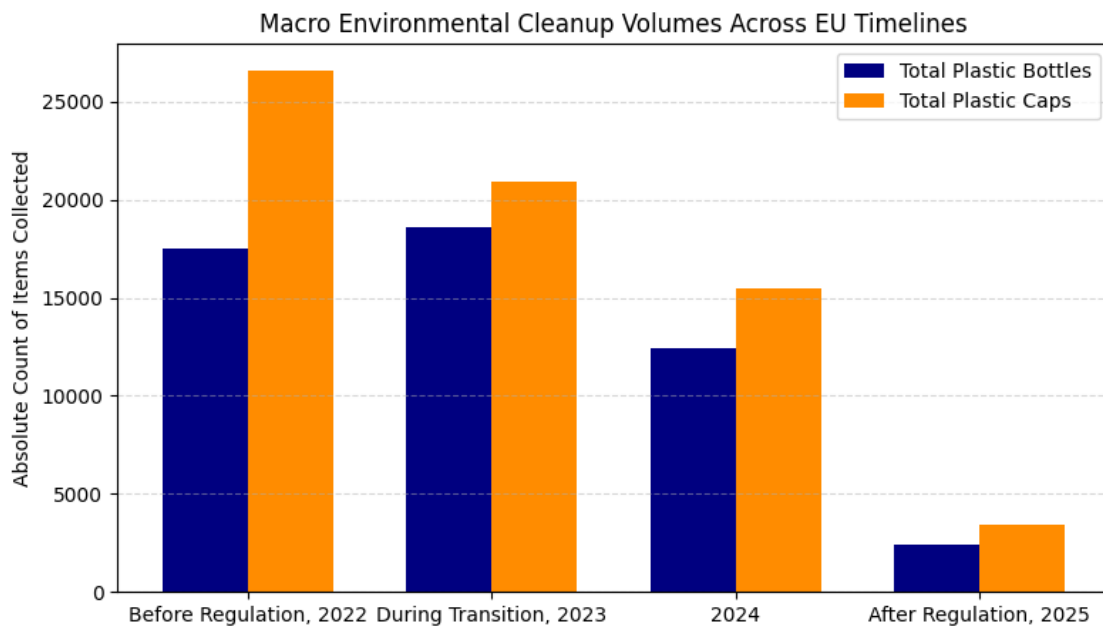
```

x = np.arange(len(categories))
width = 0.35

fig, ax = plt.subplots(figsize=(9, 5))
ax.bar(x - width/2, bottles_sum, width, label='Total Plastic Bottles',
      color='navy')
ax.bar(x + width/2, caps_sum, width, label='Total Plastic Caps',
      color='darkorange')

ax.set_ylabel('Absolute Count of Items Collected')
ax.set_title('Macro Environmental Cleanup Volumes Across EU Timelines')
ax.set_xticks(x)
ax.set_xticklabels(categories)
ax.legend()
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.show()

```



```

[84]: categories = ['Before Regulation, 2022', 'During Transition, 2023', 'After_
      Regulation, 2025']
bottles_sum = [before_1yr['Beverage bottles (plastic)'].sum(),
      during_1yr['Beverage bottles (plastic)'].sum(), after_1yr['Beverage bottles_
      (plastic)'].sum()]
caps_sum = [before_1yr['Bottle caps (plastic)'].sum(), during_1yr['Bottle caps_
      (plastic)'].sum(), after_1yr['Bottle caps (plastic)'].sum()]

```

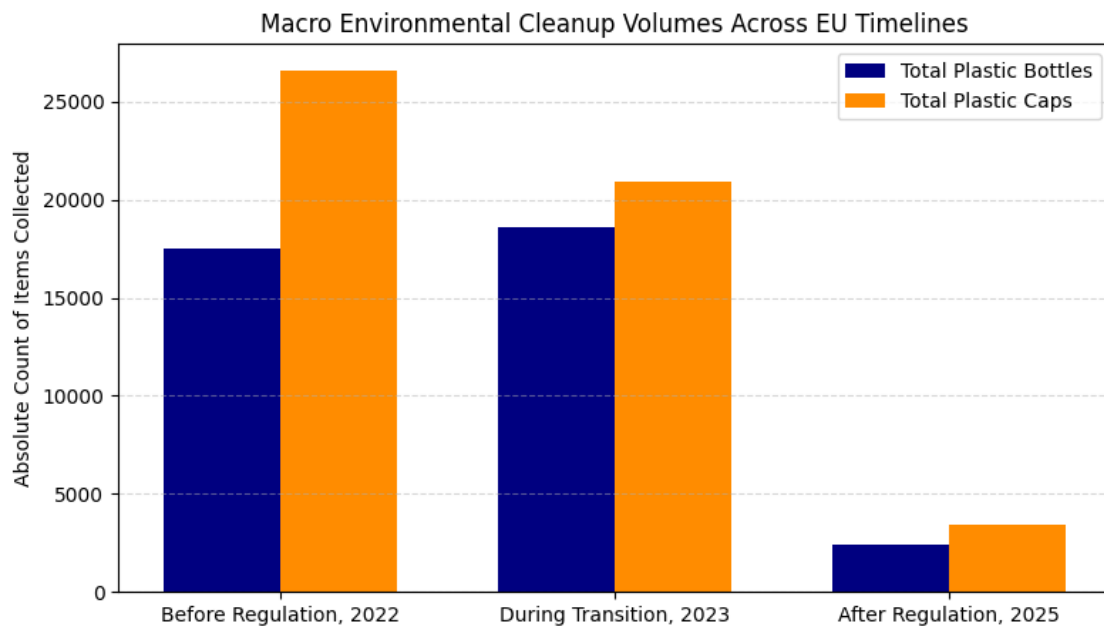
```

x = np.arange(len(categories))
width = 0.35

fig, ax = plt.subplots(figsize=(9, 5))
ax.bar(x - width/2, bottles_sum, width, label='Total Plastic Bottles',
      color='navy')
ax.bar(x + width/2, caps_sum, width, label='Total Plastic Caps',
      color='darkorange')

ax.set_ylabel('Absolute Count of Items Collected')
ax.set_title('Macro Environmental Cleanup Volumes Across EU Timelines')
ax.set_xticks(x)
ax.set_xticklabels(categories)
ax.legend()
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.show()

```



```

[85]: categories = ['Before (2021- 2022)', 'During (2023- Mid2024)', 'After(Mid2024-
      color='navy')
      color='darkorange')

macro_ratios = [before_ratio, during_ratio, after_ratio]

fig, ax = plt.subplots(figsize=(7, 4.5))

bars = ax.bar(categories, macro_ratios, color=['deepskyblue', 'orangered',
      color='navy')
      color='darkorange')

```

```

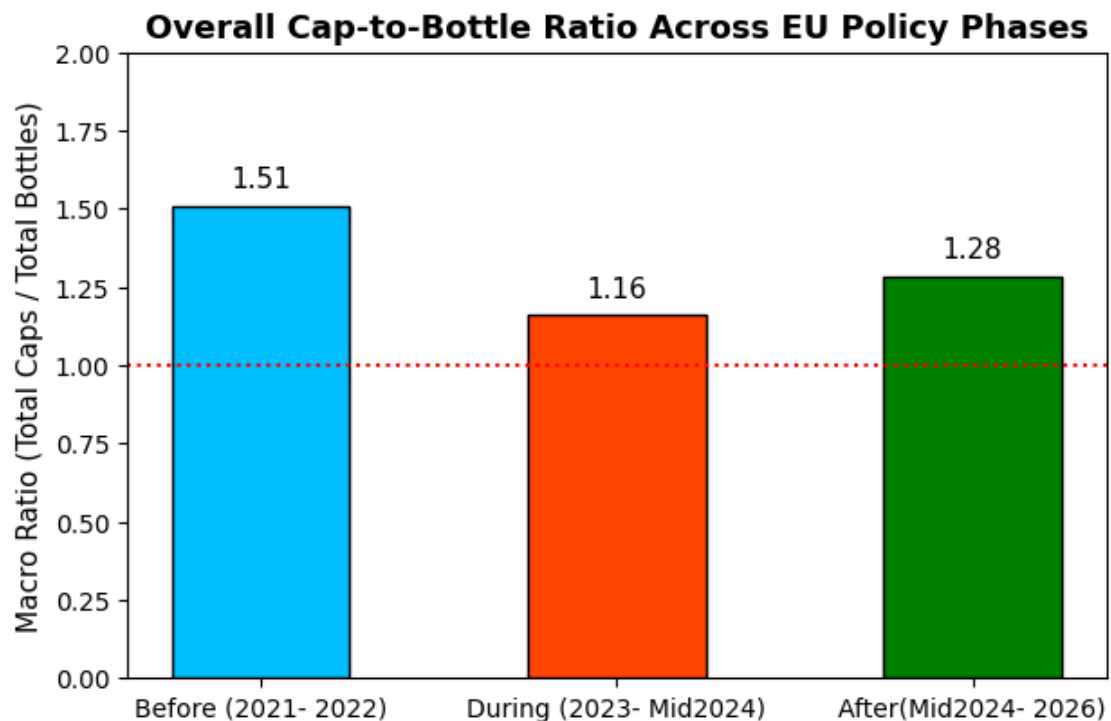
ax.set_title('Overall Cap-to-Bottle Ratio Across EU Policy Phases',
             ↪fontsize=13, fontweight='bold')
ax.set_ylabel('Macro Ratio (Total Caps / Total Bottles)', fontsize=11)
ax.set_ylim(0, 2.0)

for bar in bars:
    height = bar.get_height()
    ax.annotate(f'{height:.2f}',
                xy=(bar.get_x() + bar.get_width() / 2, height),
                xytext=(0, 5),
                textcoords="offset points",
                ha='center', va='bottom', fontsize=11)

ax.axhline(1.0, color='red', linestyle=':')

plt.show()

```



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